Accelerating cosmology

Handley Lab: KICC Welcome Day

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Accelerating cosmology from theory to data

Handley Lab: Science@KICC <wh260@cam.ac.uk>

- **Big problems**: cosmic acceleration – Λ, evolving dark energy or modified gravity? H_0, S_8 and DESI–SNe tensions: physics, systematics or priors?
- **GPUs for science**: classical Bayes, differential equations and forward models in `JAX` / `BlackJAX`.
- **AI for scientists**: agents for coding and mining legacy code for embodied intelligence. Scientists as pilots of this new scale of science.
- **The programmes**: dark energy and fundamental theory; real-time gravitational waves; LSST, Euclid and JWST readiness.



The Handley Lab (2026–2027)



Natalie Hogg
Postdoc



Matt Grayling
Postdoc



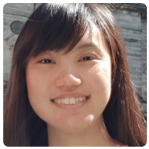
Chris Lovell
Postdoc



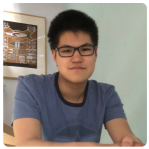
James Alvey
Kavli Senior Fellow



Nikhil Sarin
Kavli Senior Fellow



Dily Ong
PhD



Namu Kroupa
PhD



Sam Leeney
PhD



Toby Lovick
PhD



Lawrence Berry
PhD



Charlotte Priestley
PhD



Farah Wallauer
New PhD

SV

Selma Vangstein
New PhD

SH

Sam Hewson
New PhD

JL

Jaiman Lad
Part III

JGF

Jesús Girones Fraile
Part III

DP

David Parodi
Part III

MK

Manaal Khan
Part III

CM

Charlotte McCarthy
Part III

AB

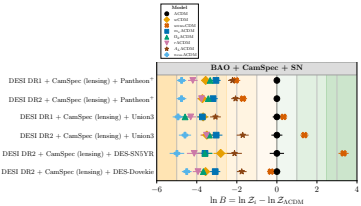
Ashwin Balakrishnan
Part III

BTB

Benedict Thomas Binu
Part III

Dark energy, tensions and fundamental theory to data

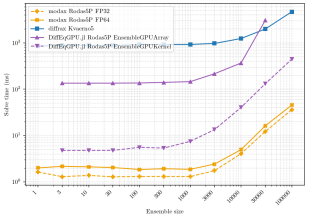
DESI DR2, GPU Boltzmann solvers and candidate Lagrangians



DESI: evolving dark energy?
Dily Ong, David Yallup, Adam Ormondroyd

Separate evolving dark energy from priors and supernova calibration. [2603.05472]

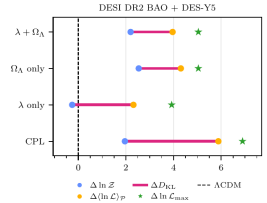
`unimpeded`: compare models and datasets using public chains. [2511.04661]



GPU Boltzmann solvers
Lawrence Berry, Oliver Hahn, Nils Schöneberg

`modax`: GPU differential equations.
`DISCO-EB`: CMB spectra without tight-coupling approximations. [2609.35661]

Next: evolve candidate theories and compare with data.



Sinah Legner: thesis, torsion models vs DESI

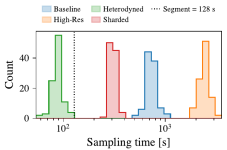
Fundamental actions to data
Sinah Legner, Wei-Ning Deng, Selma Vangstein, Sam Hewson

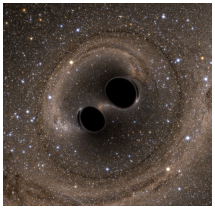
Torsion Condensation and CPT-symmetric cosmology. [2507.09228] [2509.10379]

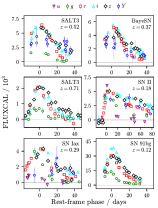
Next: agent-derived pipeline from Lagrangian to data, with full perturbations.

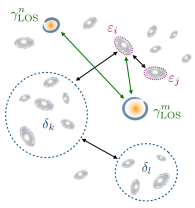
Real-time GWs and Stage-IV survey readiness

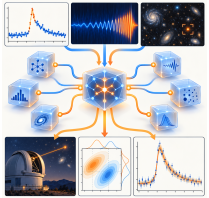
Classical Bayes for LVK, Einstein Telescope, LISA, LSST, Euclid and JWST











Real-time gravitational waves
Metha Prathaban, David Yallup, James Alvey, Nikhil Sarin, Farah Wallauer

GPU likelihoods and `SwiG` nested sampling for binary neutron-star inference. [2607.28265]

Next: Einstein Telescope, LISA, overlapping signals and shadow sirens.

Black-hole merger: SXS simulation / LIGO

Time-domain surveys and lensing
Toby Lovick, Matt Grayling, Namu Kroupa, Sam Leeney, Charlotte Priestley; Natalie Hogg, Charlotte McCarthy, Chris Lovell

ALCS: marginalise latent structure. [2603.26644] `JAX-SNANA`: simulate supernovae. [2610.19000] Quasar delays [2602.22307]; radio transients and satellite orbits.

Euclid / JWST lensing; CAMELS / FLAMELS simulations.

Agentic decision making at scale
James Alvey, Nikhil Sarin

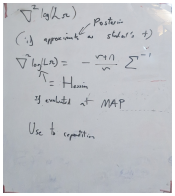
Next: transient tagging, model selection and re-fitting.

Agents: generated illustration

The Ambient AI Lab: AI for scientists, by scientists

Cambridge-owned compute, live room loops and how to join in





Local models and scientific tools
Local models and transcription on `turing` / `newton`; Cambridge UIS compute. `Alan` / `Steve`: agent tools using our shared scientific context.

Training without deskilling: students build physical intuition; senior researchers supervise agents.

Natalie Hogg: *Agentic research is oxymoronic*, Nature Astronomy. [2608.31161]

Live room loops
Toby Lovick, James Alvey, Chris Lovell, Natalie Hogg
Whiteboard capture, local transcription and equation retrieval; shared supervision context.

This term: `ai@cam` supervision hardware; Cosmos projects on ambient capture and anti-sycophancy.

People select questions, test the physics and take responsibility.

Join us: Tuesday group meeting, 14:00; Friday arXiv Scrolling Club; Friday Cavendish Gravity, 15:00 (F22); weekly Transients & GWs. `handley-lab.co.uk`